

Combining Large Language Model and Supportive Models for Legal Question Answering

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Abstract—This paper details the methods proposed by the BAMBOOVN team during their participation in the 5th Automated Legal Question Answering Competition (ALQAC 2025). During the competition we have to resolve two main tasks: Legal Document Retrieval and Legal Question Answering. For the Legal Document Retrieval task we applied an ensemble method: semantic searching with embedding outputs from BGE-M3 transformer model and traditional ranking methods using word information: BM25, TF-IDF supported by a topic classification model. After pre-ranking step we used a relevant classification model to grade the retrieved document as relevant or not relevant. These supportive models (topic/relevant classification) were trained by combining both original data and LLM-augmented data for better accuracy. The final score is combined using tuned hyperparameters and will be compared with an ensemble threshold to get filtered. For the Legal Question Answering task, we have three types of questions: true/false questions, multiple-choices question and free-text question. Suppose that the retrieved document from previous stage is correct, we address this problem by using Large Language Model (LLM) with prompt engineering technique to attain the generalization of LLM on the question answering task. While our approach did not attain the highest scores in the competition, it proved that using LLM-synthesized data within an adaptive pipeline (integrating LLMs with auxiliary models) effectively mitigates data scarcity limitations.

Index Terms—Large Language Model, Legal Document Retrieval, Legal Question Answering, Retrieval-Augmented Generation

I. INTRODUCTION

The exponential growth of legal documentation and the increasing complexity of legal frameworks have created a pressing need for efficient and accurate Legal Question Answering (LQA) systems. Such systems promise to democratize access to justice, enhance legal research efficiency, and support practitioners in navigating vast repositories of statutes, case law, and regulations. However, automating LQA poses significant challenges, including the precise retrieval of relevant documents from large corpora and the nuanced understanding required to generate correct answers to diverse legal queries (e.g., fact verification, multiple-choice, and open-ended reasoning).

Traditional approaches to legal information retrieval (IR), such as keyword-based methods (e.g., BM25, TF-IDF), offer efficiency but often struggle with semantic matching and contextual relevance. Conversely, while modern neural retriev-

ers and Large Language Models (LLMs) excel at semantic understanding, they can be computationally intensive and may lack the precision needed for highly specialized legal domains without careful tuning. Similarly, in the Question Answering (QA) phase, LLMs demonstrate remarkable generative capabilities but are prone to hallucination and inconsistency when domain-specific knowledge is inadequately grounded.

This paper presents the methodology developed by the BAMBOOVN team for the 5th Automated Legal Question Answering Competition (ALQAC 2025), addressing these dual challenges through a hybrid pipeline. For **Legal Document Retrieval**, we propose an ensemble strategy combining:

- Semantic Search using embeddings from the state-of-the-art transformer model BGE-M3 [2].
- Traditional information retrieval methods (BM25, TF-IDF) for lexical matching.
- A Relevance Classification Model, trained on both original and LLM-augmented data, to refine the ensemble's output.

For **Legal Question Answering**, we leverage prompt-engineered LLMs to generate answers across three question types (true/false, multiple-choice, free-text), conditional on high-confidence document retrieval. Our core contribution lies in synergistically integrating robust retrieval (blending semantic, lexical, and learned relevance signals) with constrained, context-aware LLM inference to enhance accuracy and reliability.

The remainder of this paper is structured as follows: Section II reviews related work in legal information retrieval and question answering. Sections III details our methodology for document retrieval and question answering, respectively. Section IV describes experimental setups and results from ALQAC 2025. Section V discusses limitations and conclusions with future directions.

II. RELATED WORK

Automatic Legal Document Retrieval (ALDR) and Legal Question Answering (LQA) represent critical subfields within legal AI, drawing significant research attention due to their practical impact on judicial efficiency and accessibility. This section surveys key advancements and methodologies relevant to our approach.

Traditional IR techniques have long been applied to Vietnam legal corpora. Keyword-based methods like TF-IDF and BM25 remain popular for their simplicity, efficiency, and interpretability in early systems like UIT-DarkCow [5], [10], [15]. NOWJ-XOSA [9] proposed a solution that combines lexical strategy with semantic searching using strong base model to achieve top 2 on retrieval task. NOWJ [6] in ALQAC 2024 has proposed an ensemble algorithm with combination of lexical matching and state-of-the-art models with data engineering steps and a tree-based model. Not participated in ALQAC Long et al [8] proposed a pipeline using lexical matching (BM25), semantic search (mBERT) and LLM-supported score. Hybrid solution has been widely applied in COLIEE 2025 (International Competition on Legal Information Extraction and Entailment) such as [11], [12] and [13]. Se7ense [14] in ALQAC 2024 combined BM25 scores with attention scores between the corpus and query, ranking based on the combined scores. To overcome the problem of lacking training data, Nguyen et al in ALQAC 2023 [4] proposed a solution that used a crawled dataset from the internet to enrich their datasets. For legal retrieval, they used an ensemble technique that combines lexical and semantic matching.

Current approaches to legal question answering bifurcate into two principal methodologies: deep learning classification systems and generative answer architectures. For span-based questions requiring exact text extraction, contemporary systems implement answer extraction techniques that directly map queries to statutory segments, as demonstrated in Nguyen et al.'s Neco@ALQAC2023 [4] framework which addresses textual grounding through contextual alignment. To overcome the pervasive challenge of lengthy legal documents exceeding conventional model context windows, chunking techniques dissect articles into semantically coherent units, with retrieval subsequently performed through lexical matching methods like BM25—an approach validated across multiple ALQAC entries. For closed-domain question types (true/false, multiple-choice), methodological divergence emerges: Nguyen et al [4] employ text-pair classification to independently score question-choice relevance, while NOWJ (ALQAC 2023) [7] implements unified BERT-based classification treating true/false as binary decisions and multiple-choice as 4-class predictions. With the raising of LLM UIT-Darkcow [5] (top 2 in ALQAC 2024), se7ense at ALQAC 2024 [14] applied prompting techniques and finetuning LLM to achieve prominent result. AIEPU [15] proposed using LLM prompting technique with voting from top-k prompt results to achieve top 1 accuracy for task 2 in ALQAC 2023.

While hybrid retrieval and LLM-based QA are established, key gaps persist:

- Most ALDR ensembles use simple score averaging based on ranking algorithm and filter threshold, lacking adaptive relevance filtering and reflexion judgement from third model (LLM/classification).
- Most solutions lack preprocessing techniques such as topic classification, rewriting question to enhance the downstream task.

- LLM-augmented data is rarely leveraged to train lightweight relevance classifiers for retrieval refinement.

Our methodology addresses these by:

- Proposing a tuned ensemble with a trainable relevance classifier (using LLM-augmented data),
- Designing an adaptive RAG pipeline for diverse legal question types while conditioning on high-precision retrieval.

III. PROPOSED METHOD

Our integrated framework for Legal Question Answering (LQA) employs a multi-stage pipeline designed to address the dual challenges of precise document retrieval and contextually grounded answer generation. The system initiates with domain-aware preprocessing, progresses through a sophisticated retrieval ensemble with learned relevance filtering, and culminates in task-specialized LLM inference. This holistic approach ensures both computational efficiency and competitive accuracy across diverse legal question types. The overall architecture is illustrated in 1.

A. Topic classification

During our research, we found out that we can preprocess the initial question to get further metadata such as the topic of the question, important keywords, etc for enhancing the retrieval pipeline's performance. Topic classification serves as the critical first stage, where incoming legal questions are categorized into specialized narrow domains (28 domains) using a fine-tuned Legal-BERT classifier. This domain identification enables two strategic advantages: (1) dynamic corpus filtering that restricts downstream document search to topic-relevant subsets of the legal database, significantly reducing computational overhead and noise, and (2) contextual priming for subsequent LLM prompts.

B. Relevant classification

The relevant classification model constitutes our novel refinement mechanism for retrieved documents. Following initial retrieval, a binary classifier [1] processes concatenated question-document pairs to predict relevance probabilities. This model serves as a filter to eliminate documents scoring below the optimized threshold, thereby ensuring that only high-confidence passages proceed to subsequent answer generation stages while significantly reducing computational overhead in the LLM processing pipeline.

C. LLM rewrite question for full text search

To address the critical limitation of traditional lexical search methods (BM25/TF-IDF) where retrieval performance is frequently compromised by unimportant keywords in natural language queries, we implement an LLM-based query refinement stage. Legal questions often contain extraneous words—common verbs, filler phrases, or contextually redundant terms—that dilute search precision when using statistical text matching algorithms. These methods inherently struggle

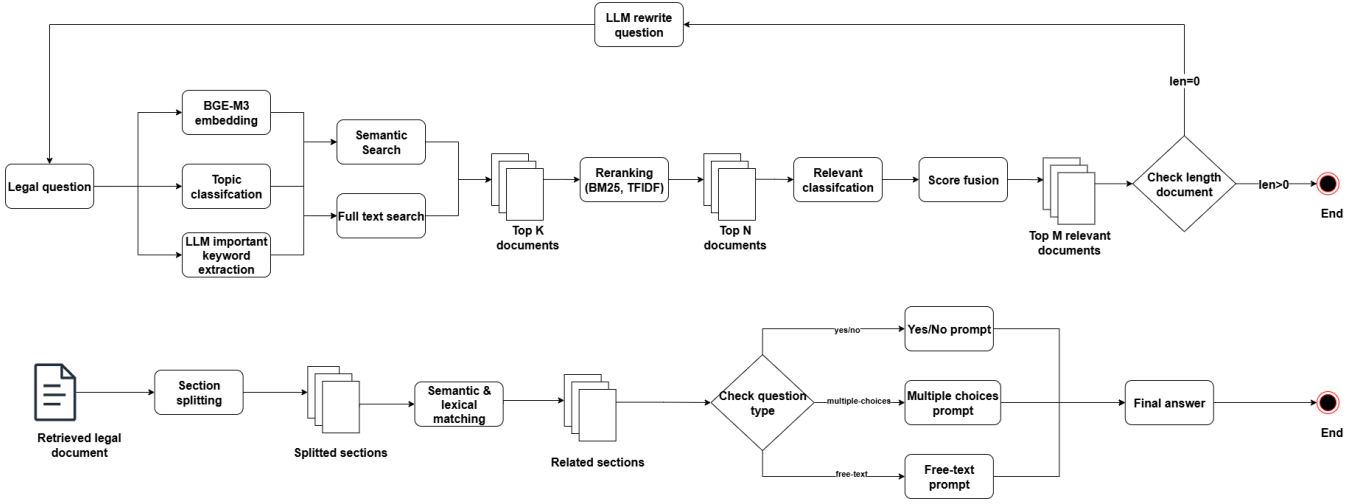


Figure 1. Overall ensemble pipeline: the upper part is the retrieval pipeline, the lower part is the question answering pipeline

Table I
LLM REWRITE QUESTION PROMPT

Type	Value
Original question	Quan hệ hôn nhân và gia đình có yếu tố nước ngoài là quan hệ hôn nhân và gia đình mà ít nhất một bên tham gia là người nước ngoài, người Việt Nam định cư ở nước ngoài, đúng hay sai?
System prompt	Bạn là một người viết lại câu hỏi, chuyển đổi câu hỏi đầu vào thành phiên bản tốt hơn được tối ưu hóa cho việc truy xuất vectorstore. Hãy xem xét đầu vào và cố gắng phân tích ý định ngữ nghĩa/ý nghĩa cơ bản.
User prompt	Câu hỏi ban đầu: {question} Chỉ trả về câu hỏi được viết lại, không giải thích gì thêm!
Rewritten question	Quan hệ hôn nhân và gia đình có yếu tố nước ngoài là gì?

to distinguish between semantically meaningful legal terminology and incidental linguistic elements, leading to irrelevant document recalls that degrade downstream performance.

D. Ensembling retrieval

Ensembling Retrieval synergizes multiple retrieval paradigms through a weighted fusion strategy. In the first stage, we execute parallel document searches: semantic retrieval via BGE-M3 embeddings [2] for semantic search, lexical matching through full text search using the important keywords extracted from step III-C. Each method returns K candidate documents. These documents are then re-ranked through an ensemble method including BM25 (optimizing term weighting), and TF-IDF (ensuring broad keyword coverage) these two methods score then fused by average. After the preranking, N highest scored documents are chosen to be scored with BERT model. The final ensemble formula score is described in formula III-D.

$$score = \alpha \times (\beta \times BM25 + \gamma \times TFIDF) + (1 - \alpha) \times BERT$$

Hyperparameters $K = 200$, $N = 10$, $\alpha = 0.5$, $\gamma = 0.5$, $\beta = 0.5$ were optimized via grid search to maximize final F_2 score. Documents falling below the empirically determined $S_{final} < 0.8$ threshold are excluded, ensuring only high-confidence passages proceed to answer generation.

E. LLM question answering

Given a related article, we know that each question only relates to a small piece of the article, our approach strategically addresses the granular nature of legal information requirements, where individual yes/no or essay questions typically reference specific segments within lengthy statutory articles, while multiple-choice questions may implicate distinct provisions across different terms of the related articles. To optimize context precision, we implement a hierarchical document decomposition protocol aligned with Vietnamese legal structure (Article $\rightarrow$ Section $\rightarrow$ Clause $\rightarrow$ Point), segmenting comprehensive articles into discrete semantic units using rule-based parsing with regular expressions. This structural normalization mitigates context window limitations while preserving juridical coherence.

The core QA framework employs Qwen2.5-7B [3], leveraging its enhanced reasoning capabilities through specialized prompt architectures (Table III). Retrieval operates at the subsection level, where:

- Semantic search (BGE-M3) and lexical search (BM25) independently identify the single most relevant legal section (top-1). If semantic search and lexical search return different sections, we put both terms to the LLM context. This technique is similar to [7] but using article's section level instead of sentence level to keep the intra-

Table II
LLM REWRITE QUESTION FOR FULL TEXT SEARCH PROMPT SETTING

Type	Value
Original question	Người đã hoàn thành cai nghiện ma túy tự nguyện đã chấp hành xong quyết định đưa vào cơ sở cai nghiện không chịu sự theo dõi hành vi sử dụng trái phép chất ma túy trong thời hạn 12 tháng kể từ ngày hoàn thành việc cai nghiện, đúng hay sai?
System prompt	Bạn là một chuyên gia về ngữ pháp tiếng Việt hoạt động trong lĩnh vực luật pháp.
User prompt	Câu hỏi của người dùng: {question}
Related article	Luật Phòng, chống ma túy article_id = 40 ... Người đã chấp hành xong quyết định áp dụng biện pháp xử lý hành chính đưa vào cơ sở cai nghiện bắt buộc chịu sự quản lý sau cai nghiện ma túy trong thời hạn 02 năm kể từ ngày chấp hành xong quyết định. ...
Full text search with original question	[]
Full text search with rewritten question	[Document(metadata="law_id": "Luật Phòng, chống ma túy", "article_id": "40")]

information of each legal section and most relevant points from both lexical and semantic queries.

- For multiple choices question, with each choice we form a query text using template $\{question\}\{choices\}$ to feed to semantic and lexical algorithm for retrieving the most related section to that choice.

IV. EXPERIMENT AND RESULTS

A. Data analysis

This year's competition dataset consists of two corpora: (1) an official task-specific collection of 82 legal questions and 3,368 legal articles spanning 28 juridical domains, and (2) a supplementary Zalo document retrieval dataset containing 3,196 queries and 61,425 legal articles.

Text length analysis (Figure 2) reveals significant lexical variation, with article lengths following a heavy-tailed distribution (range: 4–3,169 words) that necessitates specialized preprocessing. The mean article length (239.5 words) aligns with standard BERT-family model input constraints.

Question-answering queries show similar variation with a mean length of 20.66 words (min: 4, max: 90) and left-skewed dispersion, indicating frequent verbose inquiries alongside compact professional formulations.

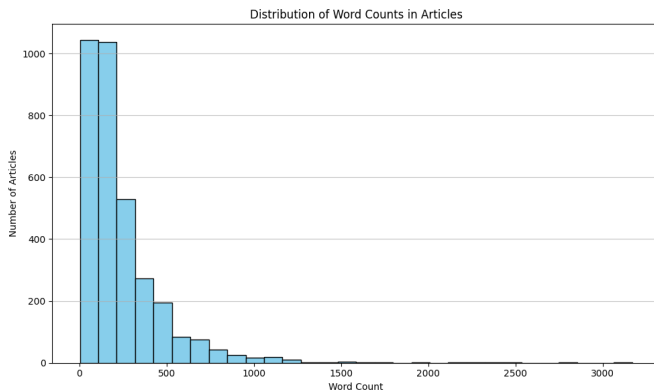


Figure 2. Legal article word count distribution

B. Finetuning classification models

For the topic classification task—which identifies the specific legal domain of a given question to optimize downstream retrieval. We curated a specialized dataset comprising:

- Original samples: 729 legally annotated questions from the ALQAC 2025 training set.
- Augmented samples from original questions: 729 synthetically generated questions from 729 original question.
- Augmented samples from original legal law: for each article of 28 topics, we used LLM prompt to generate up to 5 questions. We removed the pruned questions (empty, English/Chinese questions) and got 31,965 synthetic questions.

The combined dataset (33,423 samples) spans 28 distinct legal domains, indexed as Law IDs 0–27. Each question was mapped to its corresponding Law ID. We partitioned this data using a 90:10 train-test split (30,120 training samples, 3,303 test samples) while maintaining strict domain balance across splits.

For model development, we fine-tuned a DistilBERT transformer multilingual backbone [1], the best finetuned checkpoint achieves **92.67% accuracy** on the topic classification task. For the relevance classification, we defined a retrieved document is relevant (Class 1) or not relevant (Class 0) to a given legal question if the document can be used to answer the given question. We structured input data using a standardized template:

$\{document\} \{question\}$

This Vietnamese-language format preserves the original semantic context while providing a consistent structure for model training. The dataset construction followed three principles:

- For each original question we augmented another question using LLM, so we have 729 synthetic questions generated from 729 original questions.
- For each positive sample (relevant) we make 2 negative samples (not relevant) using legal articles from the same topic randomly.

Finally, we got a dataset of 19,654 sample with 1,997 samples in test set and 17,567 in training set using a 90:10 train-test split. The best-performing model achieved **98.8% accuracy** on

Table III
LLM PROMPT FOR QUESTION ANSWERING

Question type	Prompt type	Value
Yes/No	System prompt	Bạn là một trợ lý ảo hỗ trợ hỏi đáp pháp luật Việt Nam, bạn sẽ trả lời các câu hỏi dạng trắc nghiệm Đúng/Sai. Hãy cho điểm nhị phân 'Đúng' hoặc 'Sai', 'Đúng' có nghĩa là câu hỏi đúng dựa trên các thông tin được cung cấp.
	User prompt	Thông tin tham khảo: {context} Câu hỏi: {question} Chỉ trả về một trong hai giá trị 'Đúng' hoặc 'Sai', không giải thích gì thêm!
Multiple-choices	System prompt	Bạn là một trợ lý ảo hỗ trợ hỏi đáp pháp luật Việt Nam, bạn sẽ trả lời các câu hỏi dạng trắc nghiệm A/B/C/D. Sử dụng các đoạn thông tin đã truy xuất được để trả lời câu hỏi.
	User prompt	Thông tin tham khảo: {context} Câu hỏi: {question} {choices} Chỉ trả về một trong các giá trị A/B/C/D, không giải thích gì thêm!
Free-text	System prompt	Bạn là một trợ lý ảo hỗ trợ hỏi đáp pháp luật Việt Nam. Sử dụng các đoạn thông tin đã truy xuất được để trả lời câu hỏi. Hãy tạo ra câu trả lời ngắn gọn và súc tích gồm đầy đủ thông tin được cung cấp!
	User prompt	Thông tin tham khảo: {context} Câu hỏi: {question} Câu trả lời của bạn:

the test data. Consistent hyperparameters were applied across both fine-tuned models (DistilBERT [1]).

Embedding generation utilized the BGE-M3 multilingual model through Hugging Face Transformers, with all vector operations accelerated via GPU inference. LLM serving was handled by Ollama (v0.3.2) running Qwen2.5:7b [3] for legal question answering tasks.

This infrastructure operated on a dedicated hardware configuration featuring:

- GPU: NVIDIA GeForce RTX 3060 12GB
- CPU: Intel® Core™ i7-13700K 16 cores/24 threads 5.4GHz
- Memory: 32GB DDR5 RAM

C. Results

For rigorous evaluation across competition tasks, we implemented structured submission protocols: In Task 1 (Legal Document Retrieval), three distinct runs were generated corresponding to top-M document returns ($M = 1, 3, 5$) per query to progressively evaluate precision-recall tradeoffs. For Task 2 (Legal Question Answering), we submitted complementary runs comprising: (1) exclusive BM25-based matching, (2) hybrid BM25-semantic fusion, and (3) full-article context integration – systematically isolating the impact of retrieval granularity on answer accuracy. This tiered experimental design enabled comprehensive ablation analysis while adhering to ALQAC evaluation frameworks.

Our overall performance reveals a notable dichotomy between the effectiveness of our document retrieval system and our question-answering capability.

In the Legal Retrieval task (Table IV), our team ranked 9th out of 13 with an F2 score of **0.714**. This placement in the lower third of the ranking, with a significant performance

Table IV
LEGAL RETRIEVAL RESULT

Team	F2
Flame Reavers	0,9434765774
vund	0,89701897
UIT legalGrep	0,89701897
NOWJ	0,8766937667
UTEHY-NLU	0,8482384822
Hallucineers	0,827913279
D2	0,8238482383
UIT-Darkcow	0,7872628725
ABCD	0,7730352302
BAMBOOVN	0,7140921409
Warriors	0,6897018969
LAD	0,6588533415
VnLegalLLM25	0,5189701896

Table V
LEGAL QUESTION ANSWERING RESULT

Team	Correct	Accuracy
NOWJ	81	0,98780487800
Hallucineers	81	0,98780487800
UIT-Darkcow	81	0,98780487800
UTEHY-NLU	80	0,97560975600
FlameReavers	79	0,96341463400
vund	79	0,96341463400
BAMBOOVN	78	0,95121951200
ABCD	77	0,93902439000
D2	77	0,93902439000
VnLegalLLM25	74	0,90243902400
Warriors	67	0,81707317100
UIT legalGrep	62	0,75609756100
LAD	25	0,30487804900

gap to the top team, indicates that our retrieval pipeline was less effective at identifying the most relevant legal articles compared to the leading approaches.

Conversely, in the Legal Question Answering task (Table V), we demonstrated a much stronger performance. We achieved an accuracy of **0.951**, correctly answering 78 out of 82 questions, which earned us 7th place. Our accuracy is highly competitive, trailing the top-performing teams by a very narrow margin.

This disparity highlights a key strength in our approach: our system excels at synthesizing answers from a given set of documents, even when that set is not optimally retrieved. Our lower retrieval score implies we may have worked with a noisier or less complete set of candidate articles. However, our high answering accuracy proves that our answer generation pipeline—which leverages powerful Language Models (LLMs) and reasoning techniques—was robust enough to identify the correct solution from suboptimal retrieval results. This shows that our advanced answering component can partially compensate for deficiencies in the retrieval stage.

V. LIMITATIONS & FUTURE WORKS

Despite promising results, this work faces several limitations. The system struggles with long legal documents, and its dependence on prompting techniques with smaller LLMs (<10B parameters) increases instability and hallucination risks. Our rudimentary reranking mechanism lacks the sophistication of modern neural rankers like BGE-M3, and hardware constraints prevented full exploitation of the Zalo dataset. Future efforts will prioritize hierarchical text decomposition, advanced reasoning methods like chain-of-thought, and hybrid ranking integrating neural approaches.

CONCLUSION

This study empirically validates the efficacy of Large Language Models (LLMs) in legal-domain Retrieval-Augmented Generation (RAG) pipelines, demonstrating their dual capacity for synthetic data augmentation and operational support within knowledge-intensive workflows. Through synergistic integration of LLMs with transformer-based architectures, our resource-constrained implementation—deployed on limited hardware—achieved competitive performance in the ALQAC 2025 benchmark (top 7 in task 2). The results establish that strategically deployed LLMs can overcome computational limitations while maintaining accuracy in specialized legal contexts, offering a viable pathway for efficient legal AI systems where infrastructure scalability remains constrained.

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